

Session 10 Lab

B1.

Output:

Vehicle constructor: Toyota

Car constructor: 4 doors

Explanation: The base class constructor executes first because Car calls base(brand), then the Car constructor body runs.

B2.

Answer: No, it does not compile.

Reason: legCount is protected. Bird inherits from Animal, but it cannot access a protected member through an object of a sibling derived class (Dog). Protected members can only be accessed through the current class or its own derived instances.

B3.

Answer: Yes, it compiles.

Area = $3.14 \times 2 \times 2 = 12.56$

Output:

12.56



F:\iti\Session_10_Inheritance\ X



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=== StudentPortal Menu ===
1. Register a new student (with GPA)
2. Register a new student (no GPA yet)
3. Register a new instructor
4. Register a new admin
5. Print everyone's basic info (Person-level)
6. Print one person's full summary (by name)
7. Show total students ever created
0. Quit
Choose an option:
4
Enter the admin's full name:
youssef ezzat
Enter access level:
3
Admin registered.

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1. Register a new student (with GPA)
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0. Quit
Choose an option:
4
Enter the admin's full name:
khaled
Enter access level:
2
Admin registered.

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1. Register a new student (with GPA)
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7. Show total students ever created
0. Quit
Choose an option:
5
=== Everyone (Person-level view) ===
Person: youssef ezzat
Person: khaled

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6. Print one person's full summary (by name)
7. Show total students ever created
0. Quit
Choose an option:
6
Enter the full name to look up:
youssef ezzat
Person: youssef ezzat
Access Level : 3

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|
```